The network is now properly configured using two public subnets. Let's configure the cache next.

11.4.2 Cache: Security group, subnet group, cache cluster

You will add the ElastiCache for Redis cluster now. You learned how to describe a minimal cache cluster earlier in this chapter. This time, you'll add a few extra properties to enhance the setup. The next code listing contains the CloudFormation resources related to the cache.

Listing 11.4 CloudFormation template for Discourse: Cache

```
Resources:
  # [...]
  CacheSecurityGroup:                                          ①
    Type: 'AWS::EC2::SecurityGroup'
    Properties:
      GroupDescription: cache
      VpcId: !Ref VPC
  CacheSecurityGroupIngress:                                   ②
    Type: 'AWS::EC2::SecurityGroupIngress'
    Properties:
      GroupId: !Ref CacheSecurityGroup
      IpProtocol: tcp
      FromPort: 6379                                           ③
      ToPort: 6379
      SourceSecurityGroupId: !Ref InstanceSecurityGroup        ④
  CacheSubnetGroup:                                            ⑤
    Type: 'AWS::ElastiCache::SubnetGroup'
    Properties:
      Description: cache
      SubnetIds:
      - Ref: SubnetA
      - Ref: SubnetB
  Cache:                                                       ⑥
    Type: 'AWS::ElastiCache::CacheCluster'
    Properties:
      CacheNodeType: 'cache.t2.micro'
      CacheSubnetGroupName: !Ref CacheSubnetGroup
      Engine: redis
      EngineVersion: '6.2'                                     ⑦
      NumCacheNodes: 1
      VpcSecurityGroupIds:
      - !Ref CacheSecurityGroup
```

① The security group to control incoming and outgoing traffic to/from the cache

② To avoid a cyclic dependency, the ingress rule is split into a separate CloudFormation resource.

③ Redis runs on port 6379.